

Characterization of advanced glycation end products for biochemical studies: side chain modifications and fluorescence characteristics

Annett Schmitt^{a,*}, Johannes Schmitt^b, Gerald Münch^c, Jovana Gasic-Milencovic^a

^a *Interdisciplinary Centre of Clinical Research Leipzig, Inselstr. 22, 04103 Leipzig, Germany*

^b *NIMBUS Biotechnologie GmbH, Eilenburger Str. 4, 04317 Leipzig, Germany*

^c *Comparative Genomics Center, James Cook University, Townsville, Australia*

Received 6 August 2004

Available online 27 December 2004

Abstract

Advanced glycation end products (AGEs) are known to be involved in the pathogenesis of several diseases and therefore effects of AGEs on cells are the objective of numerous investigations. Since AGEs used in biochemical studies are usually not chemically characterized, comparison of data is difficult if not impossible. To find a suitable characterization protocol, human serum albumin was reacted with different concentrations of glucose, methyl glyoxal, and glyoxylic acid. The obtained AGEs were characterized with respect to the extent of side chain modifications (lysine and arginine), the carboxymethyl lysine and carbonyl content, and the fibrillar state. Additionally, their fluorescence and absorbance characteristics were extensively studied. Although we found significant differences in the degree of modification and in AGE-specific fluorescence when using different modifiers, the results provide important information and allow comparing AGEs derived from different modifier concentrations. The results also suggest strong conformational changes within the modified proteins. In the present paper we propose a set of parameters that is sufficient to partially characterize AGEs used for biochemical studies.

© 2004 Elsevier Inc. All rights reserved.

Keywords: Advanced glycation end products; AGE characterization; Fluorescence; Side chain modifications

The well-described nonenzymatic reaction of reducing sugars with protein side chains, lipids, or nucleic acids is referred to as glycation. Products derived from the reaction of sugars with proteins, formed during long-term incubation in complex multistep reactions and rearrangements are termed advanced glycation end products (AGEs).¹ Those AGEs are known to occur in

vivo, especially in long-living proteins such as collagen and eye-lens proteins but can also be formed in vitro by incubating sugars with proteins. In vivo, AGEs were found to contribute to the onset of several diseases such as diabetic complications [1], renal insufficiency [1], and Alzheimer's disease [2].

Although biological and biochemical effects of AGEs on cells are well described, there is usually hardly any information about the AGEs used except for the preparation procedure. Therein even basic conditions of AGE preparation such as buffers, incubation time, temperature, and the kind and concentrations of modifiers and proteins used vary considerably [3]. Therefore, comparison of experimental data from different research groups or even different experimental setups is difficult. Since AGEs are known to be extremely complex mixtures of

* Corresponding author. Fax: +49 341 9715979.

E-mail address: annett_eckert@web.de (A. Schmitt).

¹ *Abbreviations used:* AGEs, advanced glycation end products; CML, carboxymethyl lysine; 2,4-DNPH, 2,4-dinitrophenyl hydrazine; GA, glyoxylic acid; Glc, glucose; HSA, human serum albumin; MG, methyl glyoxal; NBT, *p*-nitroblue tetrazoliumchloride; PBS, phosphate-buffered saline; RAGE, receptor for AGEs; BCA, bicinchoninic acid; ELISA, enzyme-linked immunosorbent assay.

different structures among which only a few are chemically characterized, it is impossible to describe the detailed structure of AGEs. Nonetheless there are several easily accessible parameters that allow partial characterization of AGEs, used for biochemical studies, with standard laboratory equipment (see [4,5] for review). A number of colorimetric and fluorimetric methods are provided to determine parameters such as modification rate of lysine and arginine side chains, fructosamine content, and aggregation state of the modified proteins. Additionally AGE-specific fluorescence and absorbance can be obtained very easily. A more detailed but also more demanding determination of specific AGE structures such as carboxymethyl lysine (CML) or pentosidine can also be performed using HPLC [3,6], mass spectroscopy [7–9], or immunological methods [10,11].

The aim of the present work was to find a suitable experimental setup that allows a partial characterization of AGE-modified proteins used for biochemical studies, can be performed with standard laboratory equipment, and requires a minimum amount of time. HSA modified with different concentrations of glucose (Glc), methyl glyoxal (MG), and glyoxylic acid (GA) was characterized by the extent of modification, CML content, aggregation state, absorbance, and fluorescence. Fluorescence and absorbance characteristics of differently modified HSA were studied in detail. Here we suggest a set of parameters that allows a partial characterization of modified HSA and should be sufficient to compare albumin AGEs used for biochemical experiments in different laboratories or for different studies and therefore provides a basis for a deeper understanding of AGEs effects on cells.

Materials and methods

Chemicals

Chemicals were purchased from the following suppliers: Sigma (Germany), HSA (Fraction V), fluorescamine, and methyl glyoxal; Fluka (Germany), 2,4-dinitrophenylhydrazin (2,4-DNPH), *p*-nitroblue tetrazoliumchloride (NBT), 9,10-phenanthrenequinone, and Thioflavin T; Merck (Germany), glucose monohydrate and glyoxylic acid monohydrate; Roth (Germany), components for PBS, and Pierce (Germany), BCA reagents 1 and 2.

Preparation of AGEs

AGEs were prepared by incubating 4 mg/ml HSA in 0.5 M phosphate buffer, pH 8, 1 mM NaN₃, 1 mM EDTA using glucose (5, 10, 20, 50, 100, 200, 300, 500 mM) and methyl glyoxal and glyoxylic acid (each 0.3, 0.6, 1, 2, 4, 6, 10 mM) as modifiers for 4 weeks at 37 °C. Pure HSA was incubated under the same conditions and was used as a control. Buffers and all glassware were autoclaved prior

to use to inactivate proteases. All solutions were sterile-filtered prior to incubation. After the incubation AGEs and control HSA were extensively dialyzed against PBS at 4 °C, aliquoted, and stored at –20 °C. Protein concentrations were determined using the BCA assay according to the suppliers' protocol.

Determination of free amino groups using fluorescamine

Protein solution (5 µl; approx. 1 mg/ml), 100 µl 100 mM Na₂HPO₄, 45 µl aqua dest, and 50 µl fluorescamine reagent (1 mM fluorescamine in acetonitrile) were mixed and incubated for 10 min in the dark in a 96-well plate. Fluorescence was measured at excitation/emission wavelengths of 390/490 nm in a FLUORO-STAR plate reader (BMG). *N*-α-acetyl-lysine from 0 to 1.5 mM was used to determine linearity of fluorescence within the expected lysine content of the protein solution. Freshly prepared HSA solution (1 mg/ml) and PBS were used as controls for aged HSA and background fluorescence, respectively.

Determination of free arginine side chains using 9,10-phenanthrenequinone

Free arginine was assayed as described previously [12]. Briefly, 50 µl (1 mg/ml) protein samples were mixed with 150 µl 9,10-phenanthrenequinone reagent (150 µM in ethanol) and 25 µl 2 N NaOH. Protein samples without added 9,10-phenanthrenequinone were used to correct results for AGE fluorescence. Following a 3-h incubation at 60 °C, 40 µl of each sample was transferred to a 384-well plate and mixed with 40 µl 1.2 N HCl. Fluorescence of the reaction product was allowed to develop for 1 h in the dark at room temperature and then measured with a TECAN Sapphire spectrometer at excitation/emission wavelengths of 312/395 nm. *N*-α-acetyl-arginine from 0 to 0.4 mM was used to test linearity of fluorescence within the expected arginine content of the protein solution. Freshly prepared HSA solution (1 mg/ml) and PBS were used as controls for aged HSA and background fluorescence, respectively.

Determination of carbonyl content using 2,4-DNPH

Carbonyl content was assayed as described previously [3] with some minor changes. Briefly, 0.1 mg of modified protein was mixed with 0.5 ml HSA solution (4 mg/ml in PBS; serves as carrier protein) and 0.7 ml 2,4-DNPH reagent (0.1% 2,4-DNPH in 2 N HCl). After a 1.5-h incubation at room temperature 0.5 ml of a 30% trichloroacetic acid solution was added and samples were left on ice for 30 min. Precipitated protein was pelleted by centrifugation (5 min, 4 °C, 14,000g) and the pellets were washed twice with 1 ml ethanol/ethylacetate (1:1, v/v). Finally the pellets were redissolved in 0.5 ml of a 6 M

guanidine hydrochloride solution (containing 133 mM Tris, 13 mM EDTA, pH 7.4) and absorbance at 365 nm was recorded in Ultraspec 3000 Photometer (Pharmacia Biotech) using guanidine hydrochloride solution as the reference. Freshly prepared HSA solution was used as control. Carbonyl content was calculated using the extinction coefficient of $21 \text{ mM}^{-1} \text{ cm}^{-1}$ [3].

Determination of fructosamine residues using NBT

Fructosamine residues were quantified using the method of Johnson et al. [13]. Briefly, 100 μl NBT reagent (0.5 mM NBT in 0.2 M carbonate buffer, pH 10.35) was mixed with 30 μl sample (approx. 1 mg/ml) and 30 μl PBS and incubated for 1 h at 37°C. Absorbance was measured at 530 nm in a FLUORO-STAR plate reader (BMG). Freshly prepared HSA solution and PBS were used as controls. Deoxymorpholino fructose at concentrations between 0 and 1 mM containing 0.5 mg/ml HSA was used for calibration.

Determination of CML content by ELISA

Ready to use 96-well plates and solutions for CML-ELISA (kindly provided by Roche Diagnostics GmbH, Germany) were used to quantify the CML content according to the provided standard protocol (method details were previously published by Misselwitz et al. [14]). 2,2'-Azinobis(3-ethylbenzothiazoline-6-sulfonic acid) was used as detection agent. Absorbance was measured at 405 nm using a Model 550 Bio-Rad microplate reader. Fresh HSA solution (1 mg/ml) and PBS were used as controls. Ready made CML standards at concentrations between 0 and 125 ng/ml were used for calibration.

Determination of fibrillar state with Thioflavin T

The fibrillar state of modified HSA was determined as previously described [14]. Briefly, 30 μl protein solution (approx. 1 mg/ml) was mixed with 100 μl Thioflavin T reagent (10 μM Thioflavin T in 100 mM phosphate buffer, pH 7.0) and fluorescence was measured at excitation/emission wavelength of 450/490 nm in a FLUORO-STAR plate reader (BMG). Fresh HSA solution (1 mg/ml) and PBS were used as controls.

Fluorescence and UV measurements

Fluorescence and UV data were recorded with a Tecan Sapphire spectrometer in 4 mg/ml protein samples. Quenching was probed by diluting highly fluorescent samples. Because slight quenching effects were observed for HSA modified with high concentrations of MG (4–10 mM), fluorescence data were obtained in solutions of 0.2 mg/ml when necessary. Fluorescence emission wavelengths were obtained in earlier experiments with MG-

modified *N*- α -acetyl-lysine and *N*- α -acetyl-arginine and BSA-AGEs derived from Glc and MG. Emission maxima at 395 nm, 440 nm (with a shoulder at 485 nm), and 520 nm were obtained in these experiments (unpublished data). Therefore fluorescence excitation spectra for HSA-AGEs and controls were recorded at emission wavelengths of 395, 440, 485, and 520 nm with excitation wavelengths of 250–370, 250–420, 250–460, and 250–490 nm, respectively. Fluorescence spectra of samples were recorded with excitation wavelengths of 330 and 365 nm (observed in excitation wavelength scans) at 350–700 and 390–700 nm. The excitation maximum for emission at 520 nm was found to be at 380 nm but no maximum in emission spectra could be detected at 520 nm when excited at 380 nm. Therefore different wavelengths were tested for emission at 520 nm, and 485 nm was found to be a reliable excitation wavelength. The emission maximum of the HSA-AGEs was found to be slightly shifted to 530 nm compared to modified amino acids, which had a maximum at 520 nm.

UV spectra of all samples were recorded from 240 to 600 nm. Fixed wavelength data were obtained at 330, 360, and 400 nm. Samples were diluted for measurements if necessary.

Results

Determination of free amino groups using fluorescamine

Results for AGEs obtained from incubation with Glc are shown in Fig. 1. The percentage of reacted lysine was found to increase rapidly up to 200 mM Glc as modifier (50% lysine side chains modified compared to control)

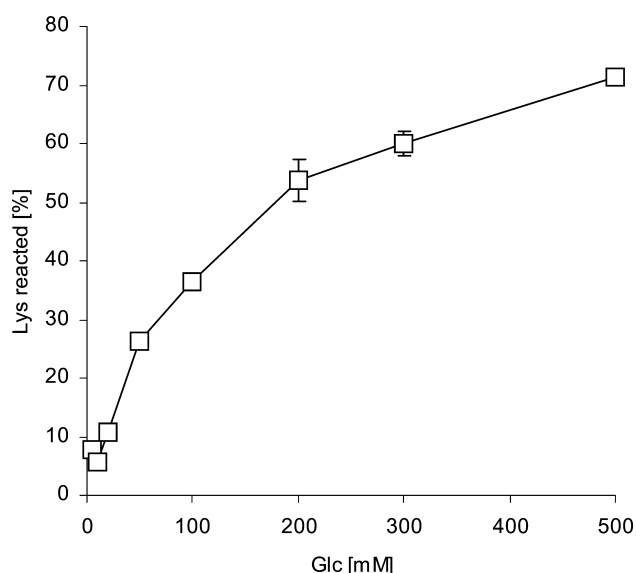


Fig. 1. Reacted lysine side chains in HSA-AGEs derived from incubation with Glc. The reacted lysine value is obtained by comparison with the lysine content of the control.

with only a slight increase between modifier concentrations from 200 to 500 mM. Even a modifier concentration as low as 5 mM Glc is sufficient to cause remarkable lysine side chain modification in HSA, which might be interesting if one is taking into account that in diabetic patients the fasting glucose value without medical treatment is higher than 7 mM Glc. The obtained level of 71% lysine side chains modified with 500 mM Glc correlates well with data from Valencia et al. [3], who found approximately 65% of BSA lysine modified after a 4-week incubation with 500 mM Glc.

For MG-modified HSA (Fig. 2) a significant degree of modification was found above 0.6 mM (6% lysine side chains modified compared to control) that increases rapidly to reach a maximum value of 83% modified lysine at 10 mM MG.

GA does not cause detectable lysine side chain modifications at low concentrations. A degree of modification

of approximately 10% was found for 10 mM GA as modifier (data not shown).

Determination of free arginine side chains using 9,10-phenanthrenequinone

Arginine side chains were found to be modified by all modifiers (Fig. 3). By far the highest degree of modification was detected for AGEs derived from MG with a maximum of 75% reacted arginine side chains. Glc and GA were also found to modify arginine to a significant extent (25% for 500 mM Glc and 19% for 10 mM GA). However, while Glc was found to modify lysine side chains to a high degree (71% with 500 mM Glc, see above), arginine side chains were found to be modified to a much lesser extent. In contrast to Glc, GA seems to modify arginine to a higher degree than lysine. MG modified lysine and arginine to about the same degree with a maximum of roughly 75% for both amino acids.

Determination of carbonyl content using 2,4-DNPH

Enhanced carbonyl contents could be detected in HSA-AGEs derived from Glc (Fig. 4) and MG (Fig. 5), whereas GA modification did not cause an increase in the carbonyl content. Glc modification resulted in a carbonyl content of 1.7 mol per mol HSA for the highest modifier concentration used. Modification of HSA with 10 mM MG yielded a carbonyl content of 8.7 mol per mol HSA. The course of the detected carbonyl contents indicates that higher modifier concentrations would result in a further significant increase in protein-bound carbonyl groups. This is especially true for HSA-AGEs derived from MG, where a linear increase in carbonyl content was found up to a modifier concentration of 10 mM.

Determination of CML content by ELISA

Results for HSA-AGEs obtained from Glc incubation are shown in Fig. 6. Although CML formation was

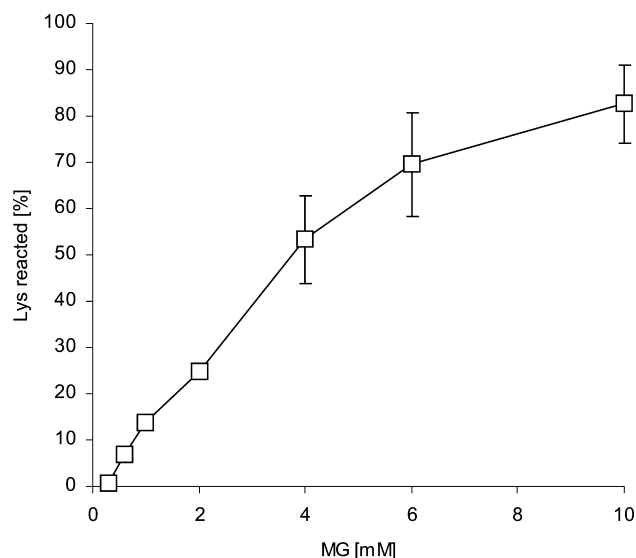


Fig. 2. Reacted lysine side chains in HSA-AGEs derived from incubation with MG. The reacted lysine value is obtained by comparison with the lysine content of the control.

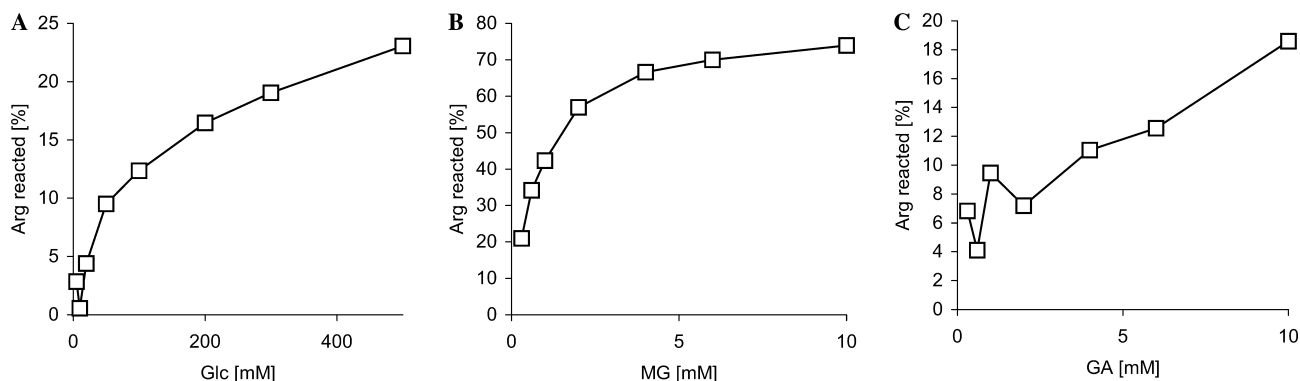


Fig. 3. Determination of free arginine side chains in HSA-AGEs derived from Glc (A), MG (B), and GA (C). Data were corrected for AGE-derived fluorescence before calculation. The reacted arginine value is obtained by comparison with the arginine content of the control.

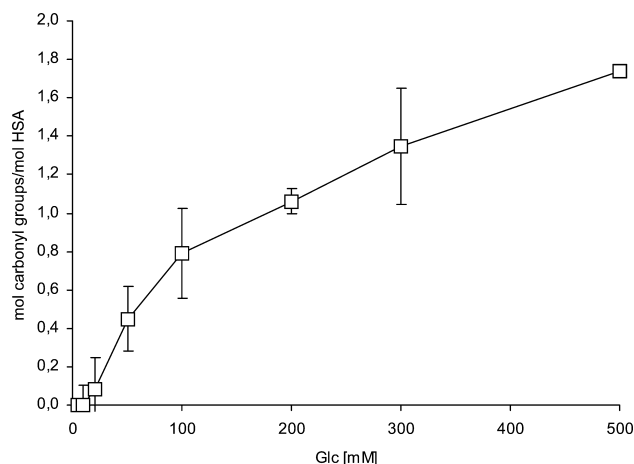


Fig. 4. Carbonyl content in HSA-AGEs derived from incubation with different Glc concentrations.

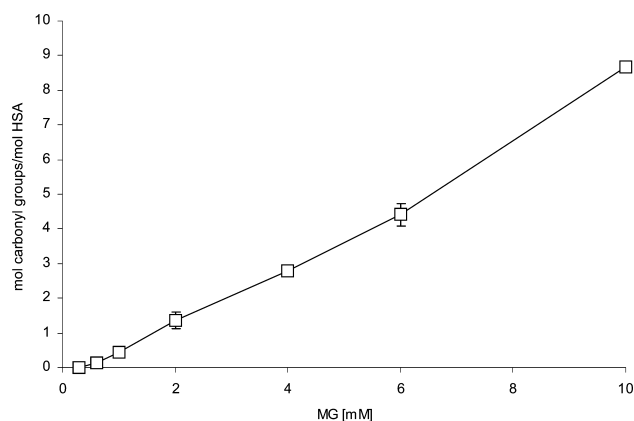


Fig. 5. Carbonyl content in HSA-AGEs derived from incubation with different MG concentrations.

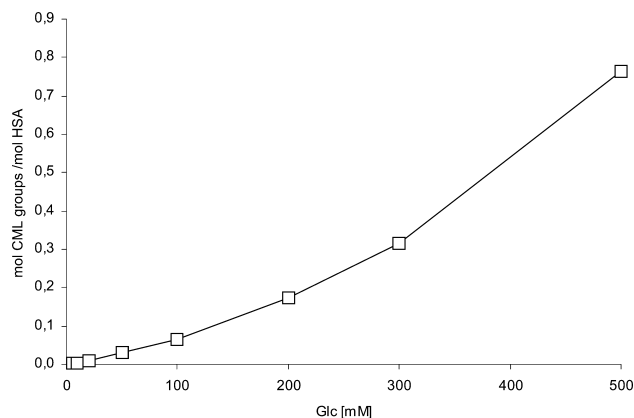


Fig. 6. CML content of HSA-AGEs derived from incubation with different Glc concentrations.

negligible at low Glc concentration, it increased gradually with modifier concentrations above 20 mM. However, the molar content of CML per HSA was very low for all Glc concentrations with a maximum CML content of approximately 0.8 mol per mol HSA. Even lower values were obtained for MG-derived AGEs (Fig. 7),

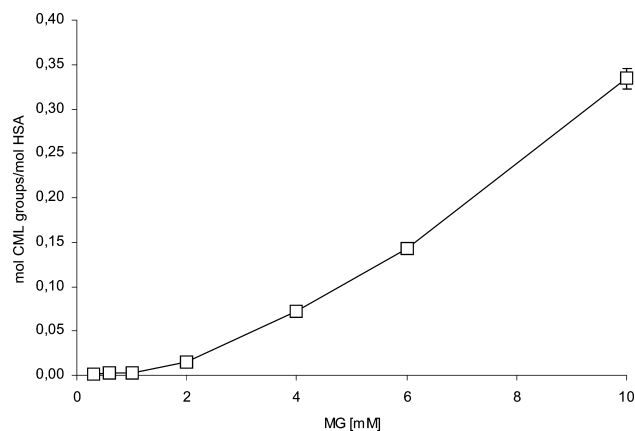


Fig. 7. CML content of HSA-AGEs derived from incubation with different MG concentrations.

yielding a CML content of approximately 0.3 mol per mol HSA. Incubation with GA at all concentrations did not yield detectable CML concentrations. CML-modified proteins are usually produced by incubating the protein with GA and NaCNBH₃ for 24 h at 37 °C (e.g., [3,6]). As no CML could be detected after a 4-week incubation with GA, the addition of a reducing agent to the incubation mixture seems to be absolutely essential for the formation of CML from GA, even at high GA concentrations and with long-term incubation.

Determination of the fibrillar state of incubated proteins with Thioflavin T

Determination of fructosamine residues using NBT

As expected, fructosamine residues were only found when HSA was modified with Glc (Fig. 8). The total number of fructosamine residues is very high especially for high modifier concentrations with a maximum value of roughly 53 residues (88% of the lysine side chains) per molecule HSA.

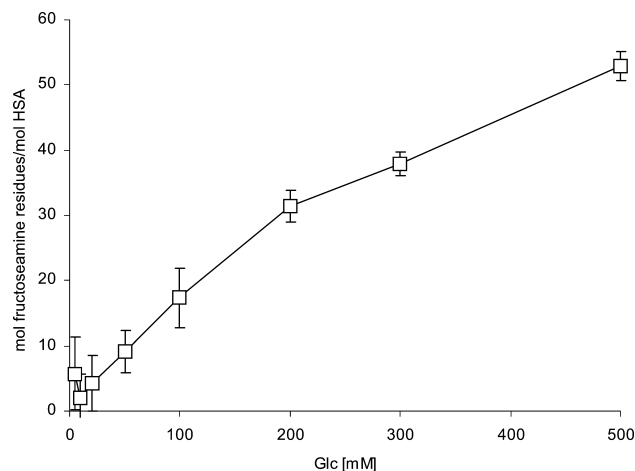


Fig. 8. Fructosamine content in HSA-AGEs derived from incubation with different Glc concentrations.

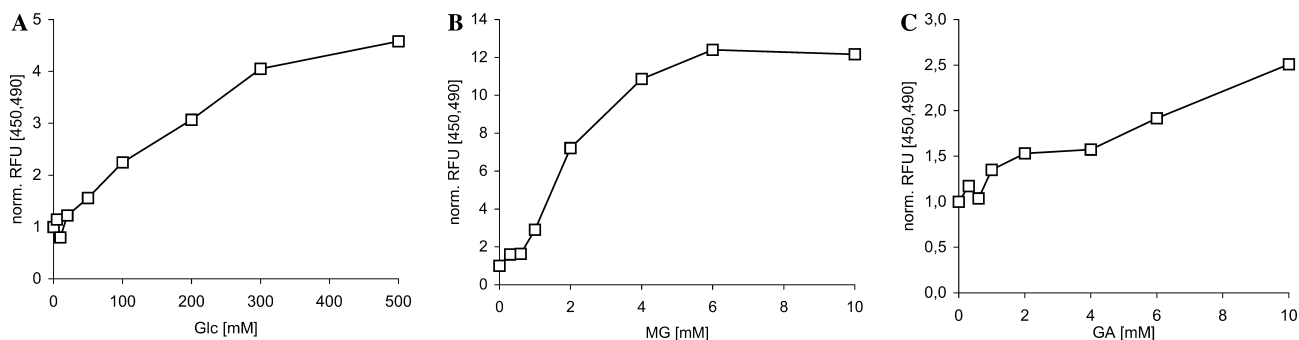


Fig. 9. Fibrillar state of HSA-AGEs derived from Glc (A), MG (B), and GA (C). All data were normalized to the fluorescence obtained with HSA control.

Thioflavin T was previously shown to interact with amyloid structures in proteins [15]. The detection of amyloid fibrils in proteins is based on the unique feature of this dye to fluoresce brightly when bound to fibrillar structures in proteins, while Thioflavin T in its free form is only weakly fluorescent [16]. As shown in Fig. 9 the fibrillar state of all incubated proteins is enhanced compared to that of the HSA control. This effect is most prominent for MG- and also very clear for Glc-modified HSA. Small but significant enhancement in Thioflavin T fluorescence was also found for GA-modified HSA, especially at high modifier concentrations. However, for GA and Glc modifications the increase in fluorescence was found to be continuous while MG incubation causes a large increase in fluorescence up to 6 mM MG with no further change at 10 mM MG. Even HSA control showed a slightly increased Thioflavin T fluorescence (1.35) compared with nonincubated HSA (1). These results strongly suggest considerable changes in the conformation of all incubated proteins. *In vivo*, such conformational changes might cause a loss of biological activity of the modified protein, e.g., a loss of enzymatic activity as described by McCarthy et al. [17] and Seidler and Kowalewski [18]. In the case of HSA, glycosylation causes significant changes in site-specific affinities of drugs (data to be published), which might cause significantly changed transport capacities and therefore affects the distribution of drugs and other small molecules *in vivo*.

Fluorescence characteristics of modified HSA

Since fluorescence data are quite simple to obtain, they are often used to detect the presence and/or formation kinetics of AGEs *in vivo* and *in vitro* [3,19]. A widely accepted assumption is that the higher the fluorescence intensity, the higher the degree of modification [4]. However, detailed fluorescence data of differently prepared AGEs are rarely published. Therefore we systematically studied the fluorescence characteristics of all HSA-AGEs and controls.

Two excitation maxima were observed in excitation spectra with emission wavelengths at 395, 440, and 485 nm. For emission at 395 nm maximum excitation was found between 325 and 335 nm, which is in agreement with data from Westwood and Thornalley [20], who reported an excitation maximum of 335 nm for emission at 406 nm for Glc-modified HSA. This fluorescence accounts for argpyrimidine and pentosidine, both well-known AGE structures derived from arginine [6]. Additionally, Lo et al. [21] reported a fluorescence with maximum excitation at 320 nm and maximum emission at 398 nm for MG-modified *N*- α -acetylarginine. The fluorescent product from this reaction was identified as *N*- α -acetyl-*N*- δ -(5-methyl-4-imidazolone-2-yl)ornithine [21]. For emission at 440 and 485 nm the excitation maximum was found between 360 and 370 nm. The fluorescence at excitation/emission wavelength of 360/430 nm was first reported for AGE-modified lens proteins by Monnier and Cerami [22] and was found to be a useful tool to determine the total AGE fluorescence [5].

To allow comparison of fluorescence data from different experiments all data are given in relation to the HSA control, which was found to show fluorescence identical to that of freshly prepared HSA. First, all data were corrected for the PBS background. Second, the spectra were normalized by dividing the spectra with the intensity at maximum fluorescence emission $\lambda_{\max}$ of the HSA control.

The fluorescence spectra recorded at an excitation wavelength of 330 nm are shown in Fig. 10. It is clear that all AGEs show detectable fluorescence. GA-modified HSA, which failed to show detectable amounts of modifications in other methods, showed a slightly increased fluorescence even when prepared at low modifier concentrations (Fig. 10C). A remarkable reaction course was found for the modification of HSA with MG (Fig. 10B). Up to modifier concentrations of 2 mM a large increase in fluorescence intensity at the emission wavelengths of 395 nm and 440 nm was observed. For higher MG concentrations the fluorescence with emission at 395 nm decreases rapidly and is lowest for HSA-AGEs produced from incubation with 10 mM MG while fluorescence with

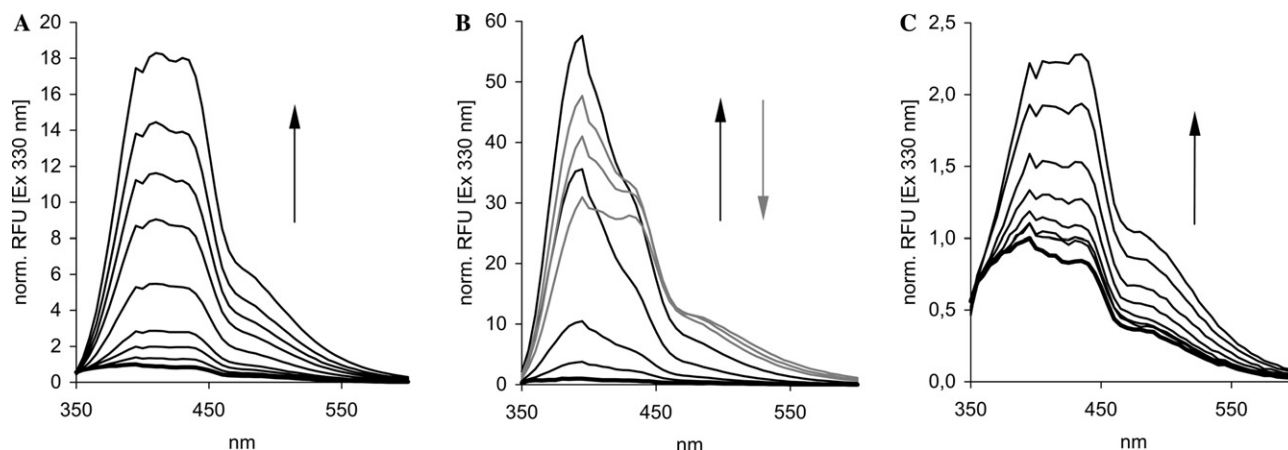


Fig. 10. Spectra recorded with excitation at 330 nm for HSA-AGEs derived from Glc (A), MG (B), and GA (C). All spectra were normalized to the spectrum of the HSA control (boldfaced line). The arrows symbolize the developing signal for increasing modifier concentrations (A,C). In (B) spectra for modifier concentrations of 0.3–2 mM are black and those for 4–10 mM are gray. Arrows in the same shades show the development of signals for increasing MG concentrations.

emission at 440 nm remains constant. The reasons for this effect are still unknown and require further investigation. However, a similar reaction course was previously observed in model reactions with α -N-terminal protected amino acids. The reaction of arginine with MG revealed the highest fluorescence for an Arg:MG ratio of 1:1 with significant decreases of the fluorescence for ratios 1:2 and 1:4 (unpublished data). These results correlate very well with the data observed here, since a MG concentration of 2 mM meets an arginine concentration of 1.6 mM in 4 mg/ml HSA solution, which is about a 1:1 ratio. This suggests the formation of different structures at different MG concentrations rather than internal quenching effects.

In contrast to the results from MG-modified HSA, for the reaction of HSA with Glc a constant increase of fluorescence with increasing Glc concentrations was discovered (Fig. 10A). Maximum fluorescence emission was found between 395 and 440 nm for all modifier concentrations.

For an excitation wavelength of 365 nm maximum fluorescence was detected at 440 nm for all AGE samples (Fig. 11). No emission peak was present at 395 nm, indicating that it occurs only with excitation at around 330 nm. Compared to the data with excitation at 330 nm, MG-modified HSA again revealed by far the highest fluorescence, but the spectra of all modified HSA were similar. However, there was again a slight loss in fluorescence for HSA modified with 10 mM MG (Fig. 11B). Fluorescence increased up to a modifying MG concentration of 6 mM and then dropped slightly to about 88% of the maximum value at 10 mM MG as modifier. Again this effect was already found in experiments with α -N-terminal-protected amino acids, where incubation with a two- and fourfold excesses of MG over arginine and lysine revealed significant decreases in fluorescence compared to fluorescence obtained from incubation of equal amounts of MG and arginine (unpublished data). For

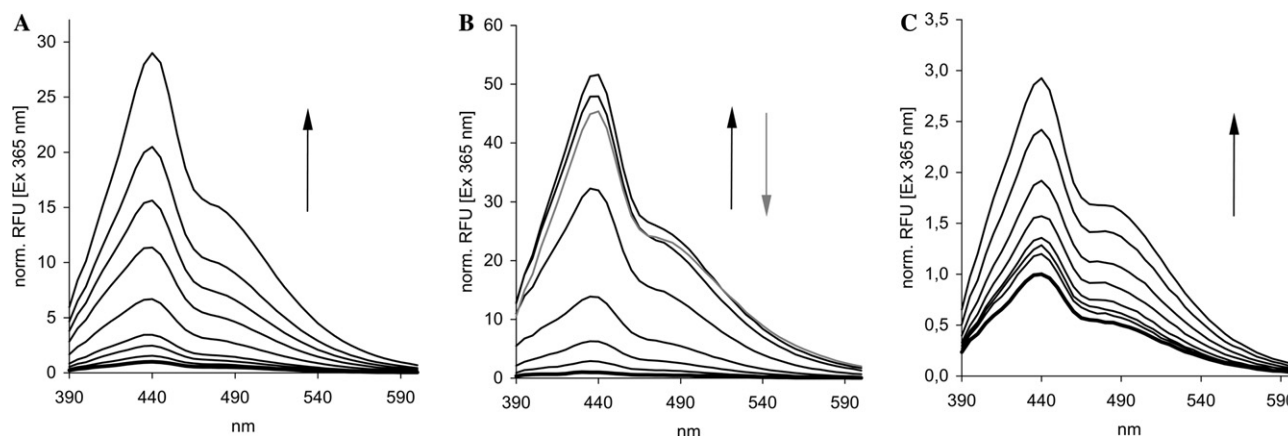


Fig. 11. Spectra recorded with excitation at 365 nm for HSA-AGEs derived from Glc (A), MG (B), and GA (C). All spectra were normalized to the spectrum of the HSA control (boldfaced line). The arrows symbolize the developing signal for increasing modifier concentrations. In (B) spectra for modifier concentrations of 0.3–6 mM are black and those for 10 mM are gray. Arrows in the same shades show the development of signals for increasing MG concentrations.

GA-modified HSA a slightly higher change in fluorescence intensity was observed at excitation/emission wavelengths of 365/440 nm compared to 330/395 nm (Fig. 11C).

However, although significant differences in the total fluorescence of AGEs derived from different modifiers were found the data indicate that fluorescence at 440 nm is correlated with the number of arginine modifications. Even though the correlations were found to be nonlinear for Glc- and MG-derived AGEs, the course of developing fluorescence follows quite well the course of arginine modifications (results not shown). Therefore fluorescence at 440 nm seems to be an indicator for progressive arginine modification when modification is derived from the same modifier. This is in agreement with the results from modified free amino acids.

A third AGE-specific fluorescence was detected in emission scans with excitation at 485 nm and maximum emission at 530 nm. Spectra obtained from emission scans are shown in Fig. 12. This fluorescence was again strongest for AGEs derived from MG and only very weak for GA-modified HSA. However, fluorescence was found to increase gradually with increasing modifier concentrations in all cases and therefore seems to be well suited to monitoring the course of AGE formation, par-

ticularly because there is almost no interfering absorbance at wavelengths above 450 nm and therefore data interpretation of fluorescence requires no additional attention to the absorbance of samples.

A further interesting aspect of the 530-nm fluorescence was found in experiments with MG-incubated amino acids, where it was clearly shown that fluorescence with maximum emission at 520–530 nm was present only for MG incubated with arginine and not with lysine. Therefore this fluorescence should be derived exclusively from arginine modifications and thus is a very useful tool to characterize AGEs.

However, the correlation between the extent of arginine modification and the fluorescence at 530 nm is not linear and the fluorescence changes are more pronounced at higher degrees of modification. The data suggest that this fluorescence is derived by arginine modifications that might be preferably formed with high modifier concentrations.

The data suggest that for highly sensitive detection of AGEs the fluorescence should be measured at all three excitation wavelengths (330, 365, and 485 nm). Emission is best detected by recording spectra as described here. But the results also demonstrate that fluorescence does not necessarily correlate with the

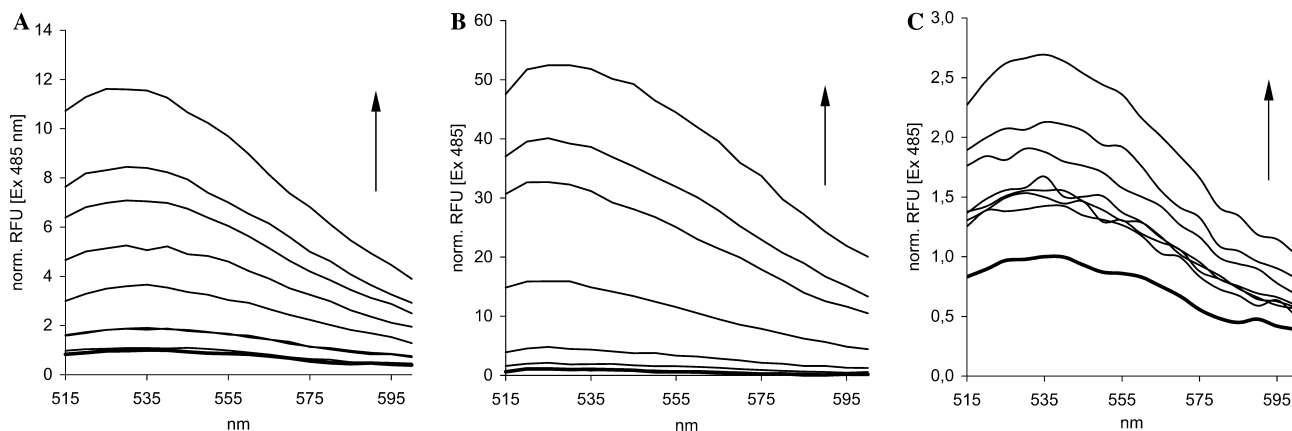


Fig. 12. Spectra recorded with excitation at 485 nm for HSA-AGEs derived from Glc (A), MG (B), and GA (C). All spectra were normalized to the spectrum of the HSA control (boldface line). The arrows symbolize the developing signal for increasing modifier concentrations.

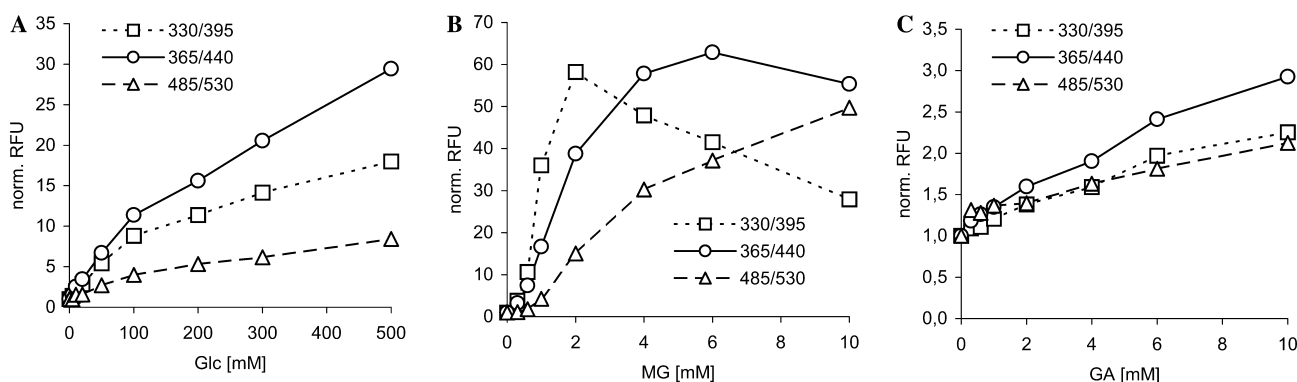


Fig. 13. Fixed wavelength data for three wavelength pairs. Fluorescence values were normalized to the corresponding HSA control fluorescence.

AGE content as was clearly demonstrated for MG-modified HSA (Fig. 13B). Therefore AGE fluorescence with emission below 500 nm provides a useful and sensitive tool to demonstrate the presence of AGEs within a protein sample but does not unconditionally allow conclusions about the total AGE content. Contrarily, fluorescence with excitation/emission at 485/530 nm was found to correlate with increasing modifier concentrations but is nevertheless very different in intensity for different modifiers.

A further interesting tool to follow the formation of AGEs is the use of the intrinsic fluorescence of proteins, namely the tryptophan and tyrosine fluorescence. As shown in Fig. 14 and Fig. 15 all modifiers decrease intrinsic protein fluorescence (excitation/emission 280/350 nm). The strongest effects were found for MG-modified HSA, where additionally a strong shift in fluorescence was observed. While the typical tryptophan fluorescence was nearly abolished to about 2% of the control protein when HSA was modified with 10 mM MG, the typical AGE fluorescence levels at 395 and

440 nm were found to increase with increasing modifier concentrations up to a concentration of 1 mM MG and then started to decline when modified with MG concentrations of 2–10 mM as already described above. A similar effect was found for the Glc-modified HSA samples although it was not as strong as that observed for MG-AGEs. The intrinsic protein fluorescence was found to decrease to about 35% when modified with 500 mM Glc. AGE-specific fluorescence was found to constantly increase with increasing Glc concentrations. GA modification solely causes a decrease in fluorescence at 350 nm at concentrations above 1 mM. But even with GA there is a remarkable loss of intrinsic fluorescence of about 35% with 10 mM GA.

The concentration-dependent decrease in tryptophan fluorescence for all HSA-AGEs is shown in Fig. 15. The decrease in intrinsic protein fluorescence was found to be correlated to the fibrillar state data obtained with Thioflavin T (Fig. 16). This can be expected since both parameters are sensitive to changes in the protein structure.

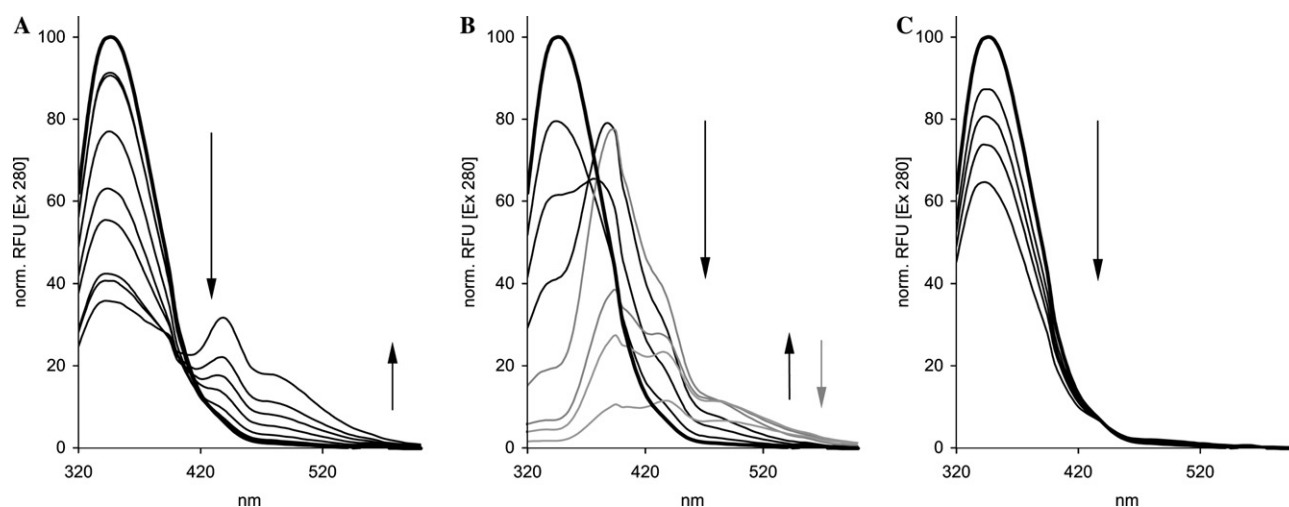


Fig. 14. Changes in intrinsic protein fluorescence caused by modifications with Glc (A), MG (B;—0.3–1 mM, —2–10 mM), and GA (C, 2–10 mM GA shown). All spectra were normalized to the spectrum of the HSA control (boldface line), where the highest value was set to 100. In A, B, and C the long arrows show the course of tryptophan-specific fluorescence with increasing modifier concentrations. The short arrows in A and B symbolize the course of AGE-specific fluorescences with increasing modifier concentrations.

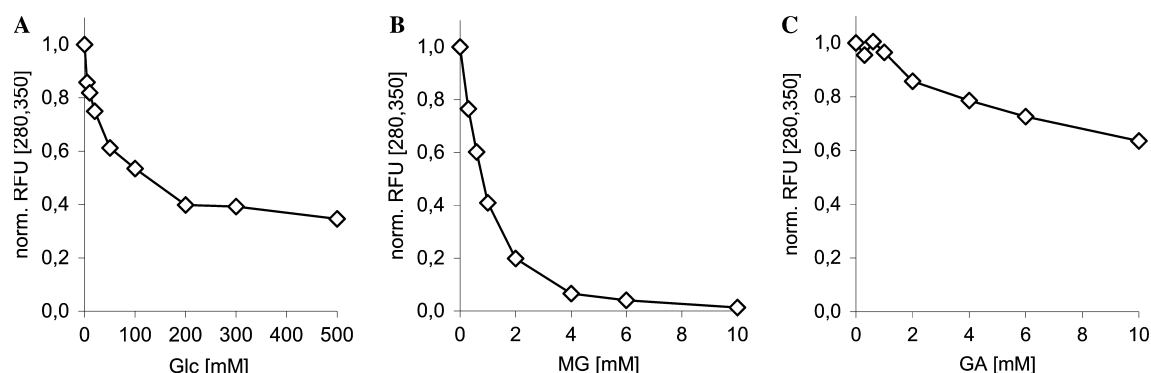


Fig. 15. Detection of tryptophan fluorescence in modified HSA. Values are normalized to the fluorescence of HSA control.

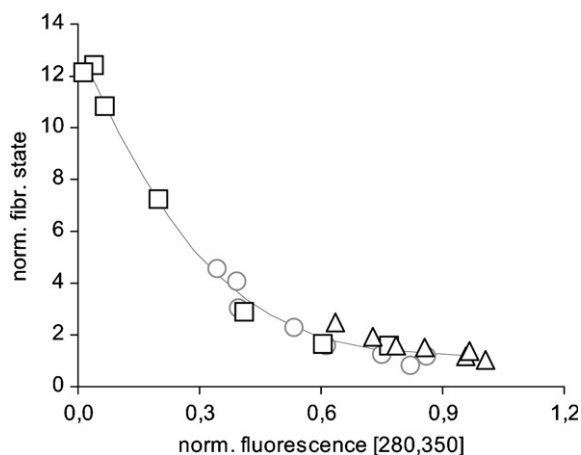


Fig. 16. Correlation between the intrinsic protein fluorescence and the fibrillar state according to the Thioflavin T test. ○, AGEs derived from Glc; □, AGEs derived from MG; Δ, AGEs derived from GA.

UV characteristics of modified HSA

UV spectra of HSA-AGEs derived from different modifiers were found to be different in some details (Fig. 17). MG modification of HSA results in a very strong increase of absorbance over the whole range of the spectrum, while GA modification results in only little changes compared to the spectrum of the HSA control. An interesting detail is the additional peak found at about 320–330 nm when HSA is modified with MG. This peak was also found in experiments with α -N-ter-

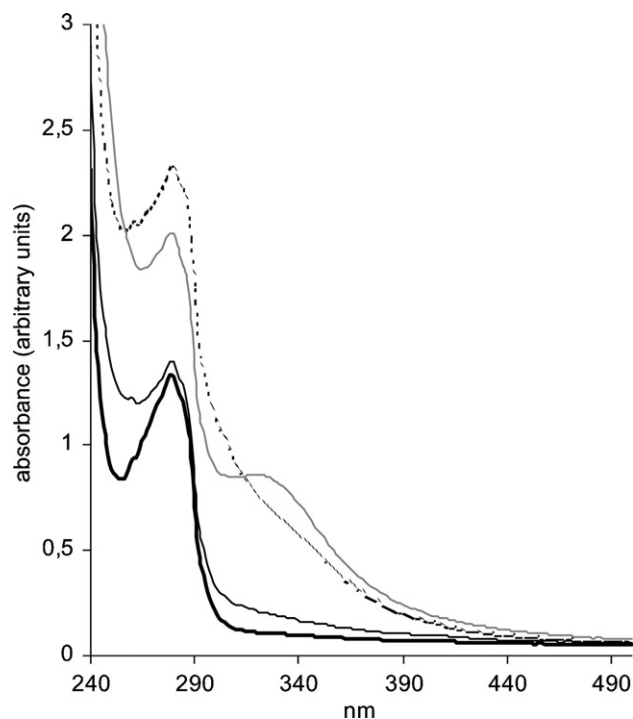


Fig. 17. Absorbance spectra of HSA-AGEs derived from --- 500 mM Glc, — 2 mM MG, — 10 mM GA, and — HSA control.

minal-protected amino acids, where a similar peak in absorbance spectrum was exclusively observed when MG was incubated with arginine (unpublished data) and should therefore be derived from modified arginine.

Fixed-wavelengths data were recorded at 330, 360, and 400 nm (Fig. 18). Due to the dependence of absorbance on concentration and optical pathlength, the measured absorbance data have to be normalized to allow comparison with absorbances obtained in different experimental setups. Because nonmodified protein does not show any absorbance above 300 nm it was not possible to normalize data in the same way as was done for the fluorescence data. We suggest additional measurement of the absorbance at 280 nm of the HSA control and use of it as reference (absorbance of incubated control was found to be equal to that of freshly prepared HSA). The measured absorbance data were corrected for the background absorbance of PBS and the plate and then normalized to the absorbance at 280 nm of the HSA control.

It was clearly shown in our experiments that the AGE absorbance increases with increasing modifier concentrations. MG-derived HSA-AGEs show the highest absorbance. GA modification yields only a very small but detectable increase in absorbance at all three wavelengths. From the absorbance data it can be concluded that absorbance between 300 and 400 nm can be used to judge the onset of AGE formation of proteins that are derived from the reaction with one modifier. From the wavelengths used in our experiments the highest sensitivity was obtained at 330 nm. Absorbance data for AGEs derived from all three modifiers were found to be in exponential correlation with the sum of lysine and arginine modifications (Fig. 19). This correlation was found to be best when absorbance data at 360 nm were used. Therefore, apparently absorbance data can be used to roughly estimate the total degree of protein modification, even though it is obvious that different structures are formed from different modifiers.

Discussion

Several methods to detect AGE structures in proteins are known. Some of these methods such as detection of the extent of side chain modifications and fluorescence or absorbance characteristics of AGEs can be performed in a short time and with standard laboratory equipment.

In this paper, we describe different methods to quantitatively assay side chain modifications and changes in protein structure and we analyzed fluorescence and absorbance characteristics of different HSA-AGEs in detail.

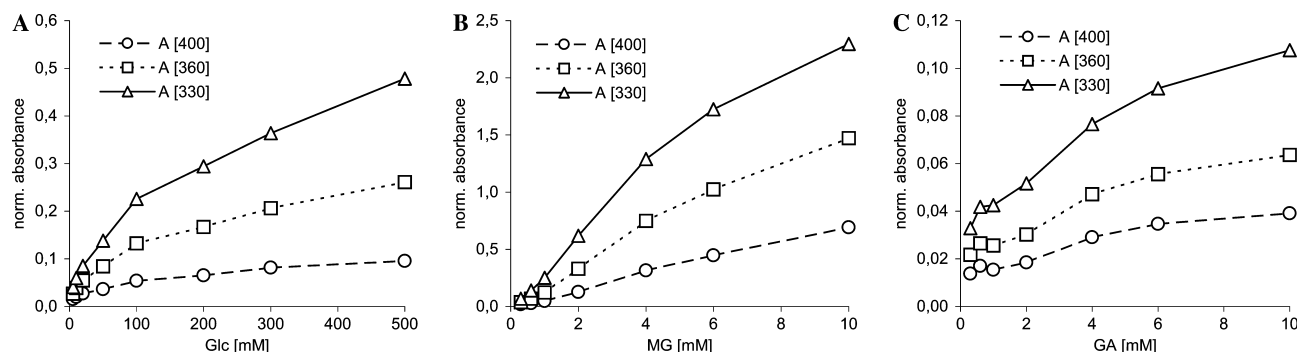


Fig. 18. Absorbance data of HSA-AGEs derived from Glc (A), MG (B), and GA (C). Absorbance data were normalized to the absorbance at 280 nm of HSA control.

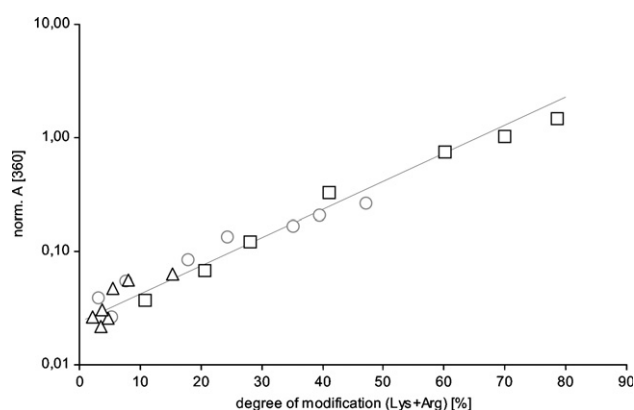


Fig. 19. Correlation between the absorbance at 360 nm and the degree of protein modification (sum of arginine and lysine modifications relative to the total number of residues). ○, AGEs derived from Glc; □, AGEs derived from MG; Δ, AGEs derived from GA.

Side chain modification data and results from the Thioflavin T assay are summarized in Table 1. All data are given either in relation to HSA control or with respect to number of modifications per HSA molecule. This allows easy comparison of data from different AGE preparations.

Especially useful is the determination of reacted lysine side chains, because it is simple to perform and highly reproducible, requires only a minimum amount of sample (5 µg protein used in the assay) and gives important information about the degree of modification in AGE-modified proteins. As shown here, lysine was modified to a high extent by Glc and MG, while modification with GA was negligible. Additionally, the extent of arginine modification should be measured for AGE characterization. Arginine was found to be modified to about 75% by 10 mM MG, while Glc (500 mM) and GA (10 mM) caused only about 20% arginine side chain modification.

Table 1
Results from side chain modification and fibrillar state experiments

	mM	Lys reacted (%)	Arg reacted (%)	Carb/HSA	CML/HSA	Fibr. state
Glc-AGEs	5	8	3	n.d.	n.d.	1.1
	10	6	1	n.d.	n.d.	0.8
	20	11	4	0.1	0.01	1.2
	50	26	10	0.5	0.03	1.6
	100	36	12	0.8	0.06	2.2
	200	54	16	1.1	0.17	3.1
	300	60	19	1.3	0.31	4.1
MG-AGEs	500	71	23	1.7	0.76	4.6
	0.3	1	21	n.d.	n.d.	1.6
	0.6	7	34	0.1	n.d.	1.6
	1	14	43	0.4	n.d.	2.9
	2	25	57	1.4	0.01	7.2
	4	53	67	2.8	0.07	10.9
	6	69	71	4.4	0.14	12.4
GA-AGEs	10	83	75	8.7	0.33	12.2
	0.3	0	7	n.d.	n.d.	1.2
	0.6	0	4	n.d.	n.d.	1.0
	1	0	10	n.d.	n.d.	1.4
	2	0	7	n.d.	n.d.	1.5
	4	0	11	n.d.	0.01	1.6
	6	3	13	n.d.	0.01	1.9
HSA control	10	12	19	0.1	0.01	2.5
	0	0	0	0	1	

A more detailed analysis of AGE-modified proteins is the detection of specific structures within the modified proteins such as the carbonyl and fructosamine content. The determination of the carbonyl content was found to be a useful tool to assess the irreversible oxidative modifications in proteins during glycation [23], while the fructosamine content is an indicator for the formation of early glycation products in the reaction with glucose [6]. In the present study carbonyl groups could be detected only in HSA-AGEs derived from MG and Glc, with a much higher carbonyl content for MG-modified proteins. Additionally, quantification of the carbonyl content can be used to determine the effectiveness of inhibitors of AGE formation as was done for Trolox C, which inhibited formation of DNPH-reactive carbonyls by 40% [23]. The number of fructosamine residues can be determined for glucose-derived AGEs even though it seems to be overestimated by this method. One reason for this effect might be that carbonyl groups were found to contribute to the determined fructosamine content [4]. Additionally, other products of AGE formation might also interfere and therefore cause overestimation of the fructosamine content. Quantification of the CML content by ELISA is also a very useful and reliable method, but unfortunately this assay is not yet commercially available. However, CML was found to be a ligand for the receptor for AGEs (RAGE) and might therefore be important for the signal-transducing pathways induced by AGEs [24,27].

Significant changes in protein structure can be detected using the fluorescent dye Thioflavin T, which binds to specific sites of amyloid fibrils [15], namely cross- β structures [25]. Formation of amyloid fibrils in bovine serum albumin glycated with glucose 6-phosphate, glucose, fructose, and glyceraldehyde/NaCNBH₃ was formerly described by Bouma et al. [25]. In their experiment they found an increase in Thioflavin T fluorescence with increasing incubation times with constant glucose 6-phosphate concentration. The presence of amyloid structures in glycated albumin might also be interesting with regard to the cellular effects of AGEs. Interestingly, many proteins that are found to bind AGEs are multiligand receptors, which also bind amyloid fibrils [25]. Both AGE and amyloid fibril binding were described for RAGE [26,27] and for different types of scavenger receptors [28,29]. Therefore, at least some of the known AGE receptors appear to be receptors for cross- β structures, which are the common motif of amyloid fibrils [25], rather than receptors for specific sugar-derived modifications of protein side chains and thus might be activated by binding amyloid fibrils rather than AGE modifications in proteins.

The second part of our study deals with the fluorescence and absorbance properties of AGEs and their possible application for AGE characterization. Results are summarized in Table 2. As already described for side chain modifications, all data are given normalized to the HSA control. Doing so allows comparison of data from different experimental setups.

Table 2
Results from fluorescence and absorbance measurements

	mM	AGE-specific fluorescence			Trp fluorescence 280/350	Absorbance		
		330/395	365/440	485/530		A [400]	A [360]	A [330]
Glc-AGEs	5	1.4	1.5	1.0	0.86	0.02	0.03	0.04
	10	2.0	2.5	1.6	0.82	0.02	0.04	0.06
	20	2.8	3.5	1.6	0.75	0.03	0.05	0.09
	50	5.4	6.7	2.7	0.61	0.04	0.08	0.14
	100	8.8	11.4	4.0	0.53	0.05	0.13	0.23
	200	11.4	15.6	5.3	0.40	0.06	0.17	0.29
	300	14.1	20.5	6.2	0.39	0.08	0.21	0.36
	500	18.0	29.4	8.4	0.35	0.10	0.26	0.48
MG-AGEs	0.3	3.8	3.2	1.1	0.77	0.02	0.04	0.07
	0.6	10.6	7.4	1.8	0.60	0.03	0.07	0.14
	1	36.0	16.6	4.3	0.41	0.05	0.12	0.25
	2	58.2	38.8	15.1	0.20	0.13	0.33	0.62
	4	47.9	57.8	30.3	0.07	0.32	0.75	1.29
	6	41.5	62.9	37.1	0.04	0.45	1.02	1.73
	10	27.9	55.4	49.7	0.01	0.69	1.47	2.30
GA-AGEs	0.3	1.1	1.2	1.3	0.96	0.01	0.02	0.03
	0.6	1.1	1.3	1.3	1.01	0.02	0.03	0.04
	1	1.2	1.3	1.4	0.97	0.02	0.03	0.04
	2	1.4	1.6	1.4	0.86	0.02	0.03	0.05
	4	1.6	1.9	1.6	0.79	0.03	0.05	0.08
	6	2.0	2.4	1.8	0.73	0.03	0.06	0.09
	10	2.3	2.9	2.1	0.64	0.04	0.06	0.11
HSA control		1	1	1	1	1 (A [280])		

AGE-specific fluorescence was previously described in several publications (e.g., [20–22,30]). It is widely used for the detection of AGEs and for following the onset of AGE formation in vitro and in vivo [3,22,31]. However, fluorescence detection in AGE samples still bears some difficulties arising from absorbance characteristics of often strongly colored AGE solutions and quenching effects. Both effects might decrease the measured fluorescence and therefore complicate the estimation of the AGE content. The influence of quenching effects can be partially circumvented by using properly diluted samples but the influence of quenching within one protein molecule and the influence of the absorbance cannot be eliminated easily. Although it is possible to correct the detected fluorescence for absorbance at excitation and emission wavelength, this would strongly complicate the evaluation of AGE fluorescence. But the results presented here clearly show that, even if quenching caused by high protein concentrations was completely excluded by using diluted samples, the detected fluorescence did not necessarily correlate with the AGE content as was shown for MG-modified HSA at excitation/emission 330/395 and 365/440 nm. However, one has to keep in mind that, especially at 330 nm but also to a lesser extent at 365 nm, the AGE samples show high absorbance, which decreases the detected fluorescence by absorbing excitation light without causing emission. Therefore, fluorescence data taken with excitation at ~330 or ~365 nm should be interpreted with care. The problem of falsifying data by high absorbance is almost completely abolished when detecting fluorescence with excitation at 485 nm, where absorbance of samples is negligible. Indeed the course of fluorescence at excitation/emission 485/530 nm for increasing modifier concentrations correlates very well with the findings from side chain modifications, where increasing modifier concentrations cause a more or less subsequent increase in modifications. Hence detection of fluorescence at 530 nm appears to be a useful tool for following AGE formation even for highly modified proteins.

In addition to AGE-specific fluorescence, intrinsic protein fluorescence can be used to characterize AGE-modified proteins. A decrease in intrinsic protein fluorescence caused by tryptophan was previously reported by Coussons et al. [32] for Glc-modified HSA. Their experiments also revealed a strong decrease in intrinsic protein fluorescence with increasing Glc concentrations. Contrarily they did not detect modification of tryptophan in amino acid analysis even after 56 days incubation with 0.5 M Glc. Therefore the decrease in tryptophan fluorescence should be due to conformational changes in the protein but again the strong absorbance contributes to the decreasing fluorescence values. On the other hand tryptophan fluorescence data correlate very well with the data from the Thioflavin T assay, where conformational changes inside the protein have already been demon-

strated. Additionally the decrease in tryptophan fluorescence is very strong, especially when comparing AGE fluorescence data for GA-modified proteins with those from tryptophan fluorescence, indicating that nonfluorescent or weakly fluorescent adducts also contribute to the loss in tryptophan fluorescence. Therefore, detection of decreasing tryptophan fluorescence could be very useful for sensitive detection especially of very weakly modified proteins or weakly fluorescent modifications. It might also be applied for the detection of nonfluorescent modifications in proteins such as CML, but this remains to be established.

Still, the easiest way to follow AGE formation was found to be the detection of absorbance of the modified samples. The spectra that were recorded from all modified samples imply that every wavelength between 300 and 400 nm is suited for detection of modifications with the highest sensitivity at the shorter wavelengths. Detection of absorbance was found to be very useful to make a rough estimation of the total degree of modification with regard to lysine and arginine modifications (Fig. 19). Best correlation of data was observed at a wavelength of 360 nm. Therefore following the absorbance at 360 nm is a simple and convincing method to estimate the onset of protein modification during incubation with sugars or sugar degradation products.

From the described results it can be concluded that for reasonable AGE characterization as many parameters as possible should be determined. As a minimum we suggest the determination of the extent of reacted lysine and arginine, the carbonyl content, the Thioflavin-T-reactive amyloid structures, and the absorbance at a wavelength of 360 nm. Additional information can be easily obtained by detecting AGE-specific fluorescence and intrinsic protein fluorescence but attention has to be paid to the limitations described above. As a basic principle all data should be given in relation to the control protein to allow comparison of AGEs used in different groups or from different preparations. Additionally utmost care should be taken when describing the AGE preparation protocol to allow reproduction of similar AGEs for biochemical studies and therefore to ease comparison of biochemical effects on cells by using similar AGEs.

Additionally the presented characterization procedures might be useful tools to estimate the effectiveness of inhibitors of AGE formation such as tenilsetam [33] or aminoguanidine [34,35] during in vitro incubation.

Acknowledgments

This work was supported by the Interdisziplinäres Zentrum für Klinische Forschung at the Faculty of Medicine of the University of Leipzig (01KS9504, Project N1) and the EU (QLK6-CT-1999-02112). The

authors thank Roche Diagnostics GmbH for providing CML-ELISA kits and Rosemarie Kientsch-Engel for helpful discussion. Additionally, we are grateful to Winnie Deuter-Conrad for revising the manuscript and Gabriele Oehme for technical assistance.

References

- [1] R. Bucala, Z. Makita, G. Vega, S. Grundy, T. Koschinsky, A. Cerami, H. Vlassara, Modification of low density lipoprotein by advanced glycation end products contributes to the dyslipidemia of diabetes and renal insufficiency, *Proc. Natl. Acad. Sci. USA* 91 (1994) 9441–9445.
- [2] M.P. Vitek, K. Bhattacharya, J.M. Glendening, E. Stopa, H. Vlassara, R. Bucala, K. Manogue, A. Cerami, Advanced glycation end products contribute to amyloidosis in Alzheimer disease, *Proc. Natl. Acad. Sci. USA* 91 (1994) 4766–4770.
- [3] J.V. Valencia, S.C. Weldon, D. Quinn, G.H. Kiers, J. DeGroot, J.M. TeKoppele, T.E. Hughes, Advanced glycation end product ligands for the receptor for advanced glycation end products: biochemical characterization and formation kinetics, *Anal. Biochem.* 324 (2004) 68–78.
- [4] A.J. Furth, Methods for assaying nonenzymatic glycosylation, *Anal. Biochem.* 175 (1988) 347–360.
- [5] S.D. Sharma, B.N. Pandey, K.P. Mishra, S. Sivakami, Amadori product and AGE formation during nonenzymatic glycosylation of bovine serum albumin in vitro, *J. Biochem. Mol. Biol. Biophys.* 6 (2002) 233–242.
- [6] N. Ahmed, O.K. Argirov, H.S. Minhas, C.A. Cordeiro, P.J. Thornalley, Assay of advanced glycation endproducts (AGEs): surveying AGEs by chromatographic assay with derivatization by 6-aminoquinolyl-N-hydroxysuccinimidyl-carbamate and application to Nepsilon-carboxymethyl-lysine- and Nepsilon-(1-carboxyethyl)lysine-modified albumin, *Biochem. J.* 364 (2002) 1–14.
- [7] A. Lapolla, D. Fedele, L. Martano, N.C. Arico', M. Garbeglio, P. Traldi, R. Seraglia, D. Favretto, Advanced glycation end products: a highly complex set of biologically relevant compounds detected by mass spectrometry, *J. Mass Spectrom.* 36 (2001) 370–378.
- [8] E. Marotta, A. Lapolla, D. Fedele, A. Senesi, R. Reitano, M. Witt, R. Seraglia, P. Traldi, Accurate mass measurements by Fourier transform mass spectrometry in the study of advanced glycation end products/peptides, *J. Mass Spectrom.* 38 (2003) 196–205.
- [9] T. Kislinger, A. Humeny, C.C. Peich, X. Zhang, T. Niwa, M. Pischetsrieder, C.M. Becker, Relative quantification of N(epsilon)-(carboxymethyl)lysine, imidazolone A, and the amadori product in glycosylated lysozyme by MALDI-TOF mass spectrometry, *J. Agric. Food Chem.* 51 (2003) 51–57.
- [10] B.K. Kilhovd, I. Giardino, P.A. Torjesen, K.I. Birkeland, T.J. Berg, P.J. Thornalley, M. Brownlee, K.F. Hanssen, Increased serum levels of the specific AGE-compound methylglyoxal-derived hydroimidazolone in patients with type 2 diabetes, *Metabolism* 52 (2003) 163–167.
- [11] M. Takeuchi, Y. Yanase, N. Matsuura, S. Yamagishi, Y. Kameda, R. Bucala, Z. Makita, Immunological detection of a novel advanced glycation end-product, *Mol. Med.* 7 (2001) 783–791.
- [12] R.E. Smith, R. MacQuarrie, A sensitive fluorometric method for the determination of arginine using 9,10-phenanthrenequinone, *Anal. Biochem.* 90 (1978) 246–255.
- [13] R.N. Johnson, P.A. Metcalf, J.R. Baker, Fructosamine: a new approach to the estimation of serum glycosylprotein. An index of diabetic control, *Clin. Chim. Acta* 127 (1983) 87–95.
- [14] J. Misselwitz, S. Franke, E. Kauf, U. John, G. Stein, Advanced glycation end products in children with chronic renal failure and type 1 diabetes, *Pediatr. Nephrol.* (2002) 316–321.
- [15] H. LeVine III, Thioflavine T interaction with synthetic Alzheimer's disease beta-amyloid peptides: detection of amyloid aggregation in solution, *Protein Sci.* 2 (1993) 404–410.
- [16] H. Naiki, K. Higuchi, K. Matsushima, A. Shimada, W.H. Chen, M. Hosokawa, T. Takeda, Fluorometric examination of tissue amyloid fibrils in murine senile amyloidosis: use of the fluorescent indicator, thioflavine T, *Lab. Invest.* 62 (1990) 768–773.
- [17] A.D. McCarthy, A.M. Cortizo, G. Gimenez Segura, L. Bruzzone, S.B. Etcheverry, Non-enzymatic glycosylation of alkaline phosphatase alters its biological properties, *Mol. Cell. Biochem.* 181 (1998) 63–69.
- [18] N.W. Seidler, C. Kowalewski, Methylglyoxal-induced glycation affects protein topography, *Arch. Biochem. Biophys.* 410 (2003) 149–154.
- [19] P. Ulrich, A. Cerami, Protein glycation, diabetes, and aging, *Recent Prog. Horm. Res.* 56 (2001) 1–21.
- [20] M.E. Westwood, P.J. Thornalley, Molecular characteristics of methylglyoxal-modified bovine and human serum albumins. Comparison with glucose-derived advanced glycation endproduct-modified serum albumins, *J. Protein Chem.* 14 (1995) 359–372.
- [21] T.W. Lo, M.E. Westwood, A.C. McLellan, T. Selwood, P.J. Thornalley, Binding and modification of proteins by methylglyoxal under physiological conditions. A kinetic and mechanistic study with N alpha-acetylarginine, N alpha-acetylcysteine, and N alpha-acetyllysine, and bovine serum albumin, *J. Biol. Chem.* 269 (1994) 32299–32305.
- [22] V.M. Monnier, A. Cerami, Nonenzymatic browning in vivo: possible process for aging of long-lived proteins, *Science* 211 (1981) 491–493.
- [23] M. Stefek, Z. Trnkova, L. Krizanova, 2,4-dinitrophenylhydrazine carbonyl assay in metal-catalysed protein glycooxidation, *Redox. Rep.* 4 (1999) 43–48.
- [24] T. Kislinger, C. Fu, B. Huber, W. Qu, A. Taguchi, S. Du Yan, M. Hofmann, S.F. Yan, M. Pischetsrieder, D. Stern, A.M. Schmidt, N(epsilon)-(carboxymethyl)lysine adducts of proteins are ligands for receptor for advanced glycation end products that activate cell signaling pathways and modulate gene expression, *J. Biol. Chem.* 274 (1999) 31740–31749.
- [25] B. Bouma, L.M. Kroon-Batenburg, Y.P. Wu, B. Brunjes, G. Posthuma, O. Kranenburg, P.G. de Groot, E.E. Voest, M.F. Gebbink, Glycation induces formation of amyloid cross-beta structure in albumin, *J. Biol. Chem.* 278 (2003) 41810–41819.
- [26] A.M. Schmidt, S.D. Yan, S.F. Yan, D.M. Stern, The multiligand receptor RAGE as a progression factor amplifying immune and inflammatory responses, *J. Clin. Invest.* 108 (2001) 949–955.
- [27] A.M. Schmidt, S.D. Yan, S.F. Yan, D.M. Stern, The biology of the receptor for advanced glycation end products and its ligands, *Biochim. Biophys. Acta* 1498 (2000) 99–111.
- [28] J. Husemann, J.D. Loike, R. Anankov, M. Febbraio, S.C. Silverstein, Scavenger receptors in neurobiology and neuropathology: their role on microglia and other cells of the nervous system, *Glia* 40 (2002) 195–205.
- [29] Y. Tamura, H. Adachi, J. Osuga, K. Ohashi, N. Yahagi, M. Sekiya, H. Okazaki, S. Tomita, Y. Iizuka, H. Shimano, R. Nagai, S. Kimura, M. Tsujimoto, S. Ishibashi, FEEL-1 and FEEL-2 are endocytic receptors for advanced glycation end products, *J. Biol. Chem.* 278 (2003) 12613–12617.
- [30] J. Leclerc, I. Birlouez-Aragon, The fluorescence of advanced Maillard products is a good indicator of lysine damage during the Maillard reaction, *J. Agric. Food Chem.* 49 (2001) 4682–4687.
- [31] R. Nagai, K. Matsumoto, X. Ling, H. Suzuki, T. Araki, S. Horiuchi, Glycolaldehyde, a reactive intermediate for advanced glycation end products, plays an important role in the generation of an active ligand for the macrophage scavenger receptor, *Diabetes* 49 (2000) 1714–1723.

- [32] P.J. Coussons, J. Jacoby, A. McKay, S.M. Kelly, N.C. Price, J.V. Hunt, Glucose modification of human serum albumin: a structural study, *Free Radic. Biol. Med.* 22 (1997) 1217–1227.
- [33] D.L. Price, P.M. Rhett, S.R. Thorpe, J.W. Baynes, Chelating activity of advanced glycation end-product inhibitors, *J. Biol. Chem.* 276 (2001) 48967–48972.
- [34] D. Edelstein, M. Brownlee, Mechanistic studies of advanced glycosylation end product inhibition by aminoguanidine, *Diabetes* 41 (1992) 26–29.
- [35] P.J. Thornalley, Use of aminoguanidine (Pimagedine) to prevent the formation of advanced glycation endproducts, *Arch. Biochem. Biophys.* 419 (2003) 31–40.